

Measurement Of Low DC Voltages Using Laptop

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We generally measure voltages with a multimeter or a voltmeter. But you can also measure low-value DC voltages up to 5V that are generally found in electronics circuits with this laptometer and see their value on laptop monitor.

For this an Arduino Uno board's serial communication function comes in handy for programming and displaying the voltages on laptop monitor using Arduino IDE and processing software. The Arduino Uno board used for measurements is shown in Fig. 1.

The Arduino Uno board is an AVR ATmega328P microcontroller based development board with 6 analogue input pins and 14 digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM, and can operate at clock frequency of 16MHz.

The board provides capability of serial communication via UART, SPI, and I2C. In this project, Arduino Uno is used for communication between

the voltage point and laptop display through serial communication. Its analogue input pin A0 is used here as input point by connecting positive terminal of voltage source to A0 pin and negative terminal to GND pin of Arduino Uno.

The simple arrangement required for voltage measurements is shown in Fig. 2. Voltage measuring probes P1 and P2 are connected across analogue pin A0 and GND pin of Arduino Uno.

When you measure voltage, the voltage received at A0 pin changes to corresponding digital value. As Arduino Uno has 10-bit ADC, the range of input analogue value is mapped as 0 to 1023, with full-scale voltage at 5V. So, for each digital level change, the voltage will be 5/1023 volts.

A part of the code written in Arduino IDE for voltage conversion is given below.

```

voltage_value = analogRead(voltage_pin);

```

```

delay(100);
voltage_V = (voltage_value * 5/1023);
Serial.println(voltage_V);

```

If 2.5V is received at analogue pin A0 as 'voltage value' in the code, its corresponding digital value will be 2.5/5 multiplied by 1023, that is, 512. And if you know the digital value (512), multiply it with 5/1023 to extract the voltage (2.5V).



Fig. 3: Layout for laptometer

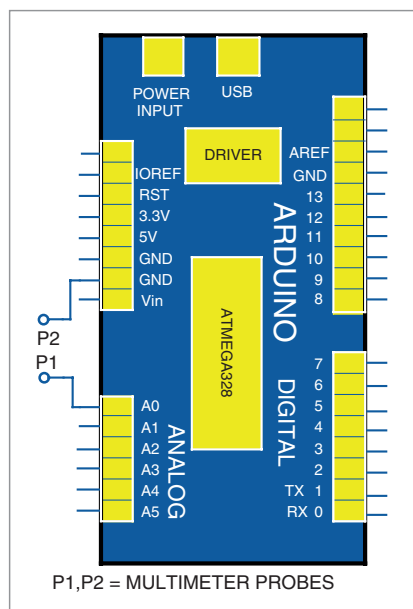


Fig. 1: Arduino Uno used for laptometer

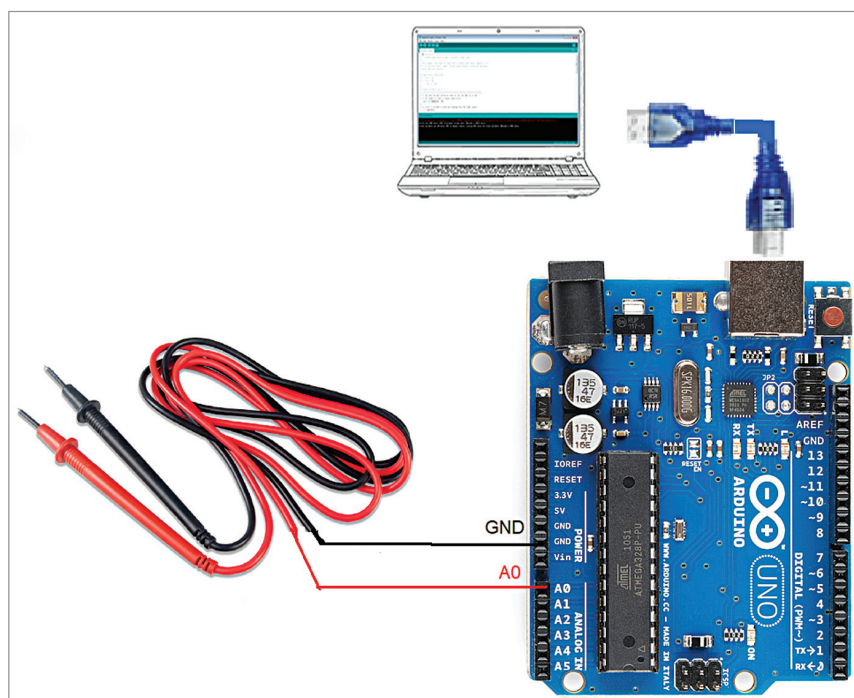


Fig. 2: Connections for laptometer